

LABORATORY WORK NO. 3

ANALYSIS OF DYNAMICAL PROPERTIES OF THE TRANSFER FUNCTION BASED ON ITS ZEROS AND POLES

The goal. To study the connection between the transient response and dynamical properties of the system based on the location of transfer function zeros and poles on the complex plane.

Theoretical background. Consider the dynamical system described by the n^{th} -order differential equation

$$y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1\dot{y} + a_0y = bg, \quad (3.1)$$

where y – the output signal, g – the input signal, $a_0, \dots, a_{n-1}, b$ – constant parameters. Here, we denote k^{th} derivative of y with respect to time as $y^{(k)}$. The roots s_k of characteristic polynomial (known as *the poles of the system*)

$$a(s) = s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0, \quad (3.2)$$

where s – the complex variable, determine the behaviour of the system step response $h(t)$ with the steady-state value $h_{\infty}(t) = k = b/a_n$, thus, also affecting such dynamical characteristics as the settling time t_s and the overshoot σ .

Defining *the cutoff frequency* as

$$\omega = \sqrt[n]{|s_1 s_2 \dots s_n|} = \sqrt[n]{a_0},$$

the characteristic polynomial (3.2) can be written as following:

$$a(s) = s^n + \alpha_{n-1}\omega s^{n-1} + \dots + \alpha_1\omega^{n-1}s + \omega^n, \quad (3.3)$$

where the coefficients $\alpha_1, \alpha_2, \dots, \alpha_{n-1}$ are calculated according to the expression

$$\alpha_k = \frac{a_k}{\omega^{n-k}}.$$

The cutoff frequency ω can be thought as the way to measure the transient response speed. That means, if in the equation (3.3) we increase ω , for example, 10 times, the transient response will be faster 10 times without any change in its behaviour. Therefore, we can consider *the normalized characteristic polynomial* as the polynomial (3.3) with $\omega = 1$, which defines *the normalized step response* $h^*(t)$ along with *the normalized settling time* t_s^* . If the overshoot of the transient response is within the requirements, the settling time t_s can be tuned by choosing the appropriate value of ω .

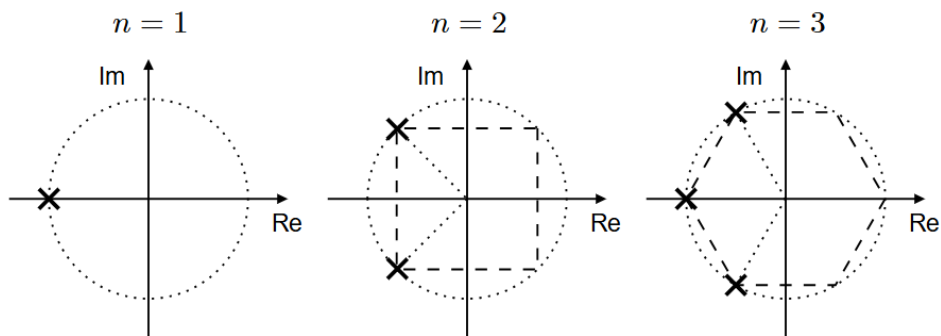


Figure 3.1 – Butterworth polynomial roots for different values of order n

In order to calculate the acceptable value of overshoot, we need to define the exact location of the characteristic polynomial's roots on the complex plane. For that, the common-used polynomials are the Butterworth polynomial and the binomial polynomial.

Butterworth polynomial is the polynomial, whose roots are located at the vertices of the regular $2n$ -gon (see Figure 3.1) with the negative real part (i.e. $\text{Re } s_k < 0$) and the same absolute value $|s_k| = \omega$. The exact expression of Butterworth polynomial is

$$a(s) = \prod_{k=1}^n \left(s - \omega e^{j\left(\frac{\pi}{2} + \frac{2k-1}{2n}\pi\right)} \right) = s^n + \alpha_{n-1}\omega s^{n-1} + \dots + \alpha_1\omega^{n-1}s + \omega^n. \quad (3.4)$$

Binomial polynomial is the polynomial with roots $s_k = -\omega$, thus, can be written as

$$a(s) = (s + \omega)^n = s^n + \alpha_{n-1}\omega s^{n-1} + \dots + \alpha_1\omega^{n-1}s + \omega^n. \quad (3.5)$$

In Table 3.1 and Table 3.2, the Butterworth polynomials and the binomial polynomials of orders first to sixth are shown respectively.

Table 3.1 – Butterworth polynomials

1	$s + \omega$
2	$s^2 + 1.414\omega s + \omega^2$
3	$s^3 + 2\omega s^2 + 2\omega^2 s + \omega^3$
4	$s^4 + 2.613\omega s^3 + 3.414\omega^2 s^2 + 2.613\omega^3 s + \omega^4$
5	$s^5 + 3.236\omega s^4 + 5.236\omega^2 s^3 + 5.236\omega^3 s^2 + 3.236\omega^4 s + \omega^5$
6	$s^6 + 3.864\omega s^5 + 7.464\omega^2 s^4 + 9.142\omega^3 s^3 + 7.464\omega^4 s^2 + 3.864\omega^5 s + \omega^6$

Table 3.2 – Binomial polynomials

1	$s + \omega$
2	$s^2 + 2\omega s + \omega^2$
3	$s^3 + 3\omega s^2 + 3\omega^2 s + \omega^3$
4	$s^4 + 4\omega s^3 + 6\omega^2 s^2 + 4\omega^3 s + \omega^4$
5	$s^5 + 5\omega s^4 + 10\omega^2 s^3 + 10\omega^3 s^2 + 5\omega^4 s + \omega^5$
6	$s^6 + 6\omega s^5 + 15\omega^2 s^4 + 20\omega^3 s^3 + 15\omega^4 s^2 + 6\omega^5 s + \omega^6$

Step responses for polynomials shown in Table 3.1 and Table 3.2 are presented in Figure 3.2 and Figure 3.3 respectively. Since both families of polynomials have roots with negative real part, the step responses are asymptotically stable. The overshoot for Butterworth polynomial is bounded: $\sigma \leq 15\%$. The overshoot for binomial polynomial is not occurring: $\sigma = 0\%$.

Standard step response method can be thought of as the method of calculating the coefficients of the system (3.1) given the parameters σ , t_s , k . Moreover, overshoot value determines the characteristic polynomial we need to choose, i.e. binomial polynomial – for zero overshoot and Butterworth polynomial – for overshoot value less than 15%.

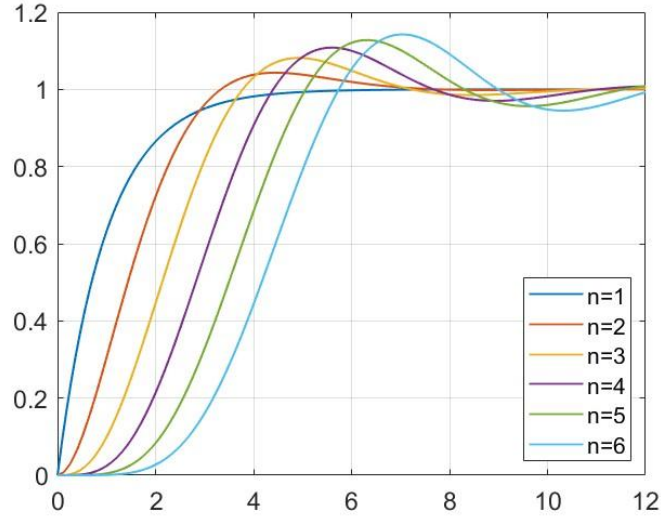


Figure 3.2. Step responses for normalized Butterworth polynomials

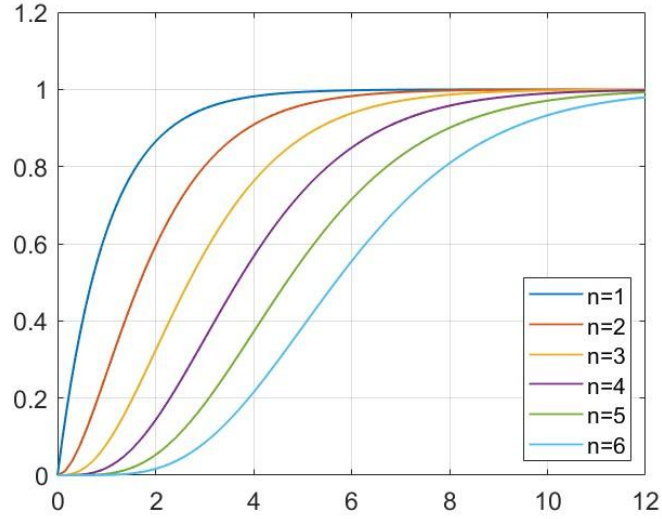


Figure 3.3. Step responses for normalized binomial polynomials

Hence, the system coefficients a_k ($k = 1, \dots, n$) can be calculated with the following algorithm:

- After choosing polynomial, calculate t_s^* of the normalized step response;
- Cutoff frequency can be calculated as $\omega = t_s^*/t_s$;
- Coefficients a_k can be calculated as $a_k = \alpha_k \omega^{n-k}$, where values of α_k can be found from Table 3.1 or Table 3.2, depending on used polynomial;
- Coefficient b can be calculated using the steady-state value as $b = ka_0$.

In some cases, you need to estimate the system performance without plotting its step response. We can define *dominant pole* of the system as a pole with the real part $\eta = \max_k(\text{Re } s_k)$, called *stability degree*. Then, if we assume that the transient response ends, when the transient response of dominant pole term ends, the settling time can be approximated as

$$t_s \approx -\frac{\ln 0.05}{\eta}. \quad (3.6)$$

The accuracy of the formula (3.6) depends on the ratio of the real part η of dominant pole to the real part of other poles of the system – the lower the value, the better will be the approximation.

Unlike the system (3.1), the step response of the system

$$y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1\dot{y} + a_0y = b_m g^{(m)} + \dots + b_0 g \quad (3.7)$$

is determined not only by the characteristic polynomial (3.2), but also by the polynomial

$$b(s) = b_m s^m + b_{m-1} s^{m-1} + \dots + b_0,$$

whose roots are called *the zeroes of the system*. By choosing the coefficients of $b(s)$, we can further decrease the settling time or guarantee the system invariant to specific input signals.

Laboratory work instructions

In the Lab Work you should take the variant as the last digit of your student number (HDU ID).

1. According to the constant values of n , t_s , k in Table 3.3 determine the parameters of the system (3.1), defined as Butterworth polynomial and as binomial polynomial. For each case, calculate the roots of characteristic polynomial (3.2) and estimate the settling time via the formula (3.6). Construct the system's block diagram in Simulink and plot step responses for both polynomials.

2. For each coefficient tuple $b_0, \dots, b_m$ presented in Table 3.4 and Table 3.5 plot the step response of the system (3.7) with coefficients $a_0, \dots, a_{n-1}$ and b , calculated in the previous task for binomial polynomial.

3. For coefficient tuple $b_0, \dots, b_m$ and input signal $g(t)$ presented in Table 3.6 plot the transient response of the system (3.7) with zero initial conditions and coefficients $a_0, \dots, a_{n-1}$ and b , calculated in the previous task for binomial polynomial. The plot must include $y(t)$ and $g(t)$.

Report requirements

1. Mathematical models of systems (3.1), (3.7) with respective block diagrams;
2. Coefficients and roots of characteristic polynomial, calculated according to the given quality parameters for both Butterworth and binomial polynomial, with the settling time estimate;
3. Numerical experiments' results (five plots of step responses and transient response for given input signal);
4. Conclusion.

Table 3.3

Variant	1	2	3	4	5	6	7	8	9	0
n	3	3	3	4	4	4	5	5	5	6
t_s	3	1	2.5	1.5	4	2	5	4	6	7
k	0.5	1.5	2	2.5	3	3.5	4	4.5	0.5	2.5

Table 3.4

Variant	1	2	3	4	5	6	7	8	9	0		
b_0	b											
b_1	0.5	1.25	1.5	2.5	1.75	2	2.25	3	2	2.5		

Table 3.5

Variant	1	2	3	4	5	6	7	8	9	0		
b_0	b											
b_1	2	2	0.5	0.5	1	0.25	0.1	0.2	0.1	0.2		
b_2	0.5	1.5	1	0.25	1.25	0.5	0.2	0.1	0.5	0.1		
b_3	0.25	1	1	1.25	0.25	0.75	0.5	0.2	0.2	0.2		
b_4	-	-	-	2	2.5	3	0.3	0.5	0.25	0.3		
b_5	-	-	-	-	-	-	3.5	4	0.25	0.5		
b_6	-	-	-	-	-	-	-	-	-	2		

Table 3.6

Variant	b_0	b_1	b_2	$g(t)$
1	1	0	0.25	$\sin 2t$
2	1	0	1	$2 \sin t$
3	0.25	0	1	$\sin 0.5t$
4	2.25	0	2	$\sin 1.5t$
5	8	0	0.5	$3 \sin 4t$
6	4.5	0	0.5	$2 \sin 3t$
7	1	0	0.25	$\cos 2t$
8	1	0	1	$3 \cos t$
9	0.25	0	1	$\cos 0.5t$
0	2.25	0	2	$\cos 1.5t$